Adaptive assessment of early vocabulary: Insights from English and Polish CAT-CDIs validation studies

Karolina Muszyńska[1], Grzegorz Krajewski[1], Magdalena Krysztofiak[1], George Kachergis[2], Virginia A. Marchman[3], & Michael C. Frank[3]

[1] Faculty of Psychology, University of Warsaw, Poland

[2] Cognition, Behavior, and Information (CBI), Arizona State University, USA

[3] Department of Psychology, Stanford University, USA

Author Note

The authors made the following contributions. Karolina Muszyńska: Conceptualization, Data Curation, Formal Analysis, Funding Acquisiton, Investigation, Methodology, Project administration, Software, Supervision, Validation, Visualization, Writing - Original Draft Preparation, Writing - Review & Editing; Grzegorz Krajewski: Conceptualization, Data Curation, Investigation, Methodology, Project administration, Software, Supervision, Writing - Review & Editing; Magdalena Krysztofiak: Conceptualization, Data Curation, Funding Acquisiton, Investigation, Methodology, Supervision, Visualization, Writing - Review & Editing; George Kachergis: Conceptualization, Data Curation, Investigation, Methodology, Software, Supervision, Writing - Review & Editing; Virginia A. Marchman: Conceptualization, Supervision, Writing - Review & Editing, Supervision; Michael C. Frank: Conceptualization, Supervision, Project administration, Writing - Review & Editing.

Correspondence concerning this article should be addressed to Karolina Muszyńska, University of Warsaw, Faculty of Psychology, Stefana Banacha 2D, 02-097 Warsaw, Poland. E-mail: karolina.muszynska@psych.uw.edu.pl

Abstract

The MacArthur-Bates Communicative Development Inventories (CDIs) are widely used parent-report tools for assessing children's early vocabulary and language development. Although CDIs provide detailed insights across a wide range of lexical and semantic categories, their length can make them impractical for screening or clinical use. Recent efforts to shorten CDIs have applied Item Response Theory (IRT) and Computerized Adaptive Testing (CAT). This study evaluates the performance of adaptive CDI instruments in English and Polish via a joint analysis of data from norming studies – some of which has been previously reported. The English CAT-CDI – covering production across the full developmental range (12–36 months) – prioritizes short administration while the Polish CAT-CDIs retain age-specific versions and emphasize higher precision. We compare reliability and validity evidence for each instrument. Cross-linguistic comparisons further expore how differences in calibration samples and linguistic structure influence item parameters. By comparing the two CAT-CDIs, we illustrate how different design choices – such as calibration sample composition and stopping criteria – can influence test outcomes, and how these insights can guide future CAT-CDI adaptations in selecting approaches best suited to their research or clinical goals.

Keywords: MacArthur-Bates Communicative Development Inventories, early vocabulary knowledge, Item Response Theory, Computerized Adaptive Testing

Word count: X

Adaptive assessment of early vocabulary: Insights from English and Polish CAT-CDIs validation studies

Introduction

The MacArthur-Bates Communicative Development Inventories (CDI) are parent-report forms that are used worldwide in research and clinical practice to measure children's language abilities, especially word knowledge. There are over 100 language adaptations of the CDI, including Polish (Smoczyńska et al., 2015), and up to three age versions in each language: Words and Gestures (CDI:WG) for children up to the age of a year and a half, Words and Sentences (CDI:WS) for older children up to around three, and CDI-III for children from three to three and a half or four (exact age ranges vary between adaptations). All CDIs include a vocabulary checklist, a list of words in which parents or other caregivers mark the words that their child "understands" and/or "understands and says." CDIs and similar parental reports draw on caregivers' extensive experiences with their child in a variety of contexts. They demonstrate adequate validity with spontaneous language samples (e.g., Gatt et al., 2015; O' Toole & Fletcher, 2010) and other direct language measures (e.g. Dale, 1991; Lemhöfer & Broersma, 2011; Smolík & Bytešníková, 2021), both in typically-developing children as well as children with an elevated likelihood or diagnosis of ASD (Belteki et al., 2025).

Because CDIs are inventories, they cover a broad range of words from many semantic and lexical categories, differing in frequency, age-appropriateness, and other properties (e.g., Frank et al., 2021; Garmann et al., 2019). This richness is a clear strength in research settings, where detailed, item-level analyses of children's vocabularies are often the goal. For practical purposes, however - such as early screening for language delay - a lengthy inventory can be less than ideal. Caregivers may need 20–30 minutes to complete the forms, which can make the task burdensome when only a single overall score is needed. For this reason, there have been many attempts to shorten the CDI word lists. The initial attempts were to simply identify a subset of items from the original lists that correlated most strongly with responses on the full list (Fenson et al., 2000; but later also Ezeizabarrena et al., 2024; Kim et al., 2014; Lasorsa et al., 2021; Smolík & Bytešníková, 2021). More recently, Mayor and Mani (2019) administered a random subset of words from the full CDI, combining these with vocabulary data from the WordBank database, and then incorporated information about the child's age, sex, and language status into Bayesian models that predicted the child's overall vocabulary size as it would appear on the full CDI. This method provided a reliable estimate of the child's language development level without requiring the complete inventory: using a 25-item lists, CDIs correlated with the full CDIs at r = .96, with an SE of .14 and a reliability of .98; using a 50-items CDIs showed slightly lower correlation at r = .94 with similar SE of .14 and a reliability of .98. However, as noted by Kachergis et al. (2022), while using age and sex to predict vocabulary improves precision with limited data, these factors can also bias individual assessments by pulling scores toward typical expectations for those demographics.

Other attempts to shorten the CDIs have incorporated Item Response Theory (IRT) and Computerized Adaptive Testing (CAT) (e.g., Chai et al., 2020; Kachergis et al., 2022; Makransky et al., 2016). IRT is a modern psychometric approach, which unlike Classical Test Theory (CTT), examines each item response and how it contributes to estimating a measured ability. This ability (often denoted as theta) is represented on a continuum: negative values correspond to lower ability, values near zero to average ability, and positive values to higher ability. A key feature of IRT is its focus on item characteristics (or parameters). In a two-parameter model, each item is described by its difficulty (the ability level required to answer correctly) and discrimination (how well the item distinguishes between individuals of similar ability levels). Once item parameters are known, every response provides information about an individual's ability. In the CDI context, IRT model captures the difficulty and discrimination of vocabulary items and the relation between parental responses to those items and the child's vocabulary knowledge, typically using large datasets such as norming samples.

Once an IRT model is established and item parameters are estimated, a computerized adaptive task (CAT) can be constructed. This involves specifying how the task should begin, how the ability is to be estimated, and the criteria for ending the task. In a CAT-CDI, items are presented to the parent one at a time. After each response, the algorithm updates the estimate of the child's vocabulary ability and selects the next item to further refine this estimate. The process continues until either a predetermined number of items is reached or the estimate achieves sufficient precision. IRT and CAT-based tasks are increasingly being developed in many areas of language assessment, including reading Ma et al. (2025), direct vocabulary testing in older children (Bohn et al., 2024, 2025) and measuring overall variability in children's learning and development Frank et al. (2025).

The first IRT-based CAT-CDI was developed by Makransky et al. (2016) who applied IRT models to normative data from the American English CDI:WS. Using the full CDI dataset, they simulated CAT-CDIs of different lengths by sequentially selecting items and filling in responses based on participants' full CDI data. Then they compared the full CDI scores with the scores from various fixed-length CATs ranging from 5 to 400 items. Notably, a CAT with just 50 items produced scores that correlated strongly with the full CDI, achieving a correlation of $r = .95$. Building on this, Kachergis et al. (2022) developed online CATs for assessing vocabulary in both American English and Mexican Spanish and extended the adaptive assessment framework to both receptive and productive vocabulary. Their CAT-CDIs were found to accurately estimate vocabulary comprehension and production using as few as 25-50 items, significantly reducing assessment time while maintaining high validity.

These developments paved the way for the creation of a CAT-CDI in Polish (Krajewski et al., in preparation). While a detailed description of the development process for both CAT-CDIs is beyond the scope of this paper, we briefly highlight key decisions that researchers must consider when developing a CAT-CDI in a new language. To illustrate, we draw on the English and Polish CAT-CDIs as examples. The primary aim of the present paper is to evaluate the performance of these two instruments in validation studies conducted in the USA and Poland. We believe that such an evaluation can guide future CAT-CDI development, enabling researchers to make more informed and strategic decisions.

Item parameters estimation. Both the English (Kachergis et al., 2022) and Polish (Krajewski et al., in preparation) CAT-CDIs used a two-parameter logistic (2PL) model, in which each item is defined by two characteristics: difficulty and discrimination. These parameters were estimated from large datasets: Wordbank data for English and CDI:WS norming data for Polish. A central design decision concerns how many CAT-CDIs to develop and what data to use for item parameters calibration. For English, two instruments were created: one for comprehension and one for production. Unlike the original CDIs, they are not divided into age-based versions. Instead, both CAT-CDIs span the entire developmental range of the CDI instruments, from the earliest CDI:WG through the CDI-III. This design reflects the assumption that early vocabulary can be captured by two latent traits - comprehension and production - and it maximizes the dataset available for parameter estimation. A practical benefit is that practitioners only need to choose between comprehension and production, without worrying about which age-specific version to administer. In contrast, the Polish CAT-CDIs retain the original structure, with separate adaptive instruments for each age version (e.g., CAT-CDI:WG for word comprehension or production for children aged 8-18 months, CAT-CDI:WS for word production for children aged 18-36 months). This may benefit practitioners who are accustomed to the age-based Polish CDI forms, allowing them to select the appropriate version based on the child's age. It may also facilitate comparison between CAT-CDI:WS and the original CDI:WS, ensuring consistency in application and interpretation.

This structural difference affects how item characteristics are estimated. The English item parameters were calibrated on data from children aged 12–36 months (Kachergis et al., 2022). The Polish CAT-CDI:WS item parameters were calibrated on children aged 18–36 months, consistent with the Polish CDI:WS norming data. As a result, an item may appear more difficult in English CAT-CDI (relative to the Polish CAT-CDI) simply because its parameters were estimated using responses from a broader and more heterogeneous sample, rather than reflecting any intrinsic cross-linguistic difference. Such variation is expected and appropriate, as each CAT-CDI is optimized for the population it was designed to assess.

At the same time, we can still expect cross-linguistic similarities. These can be found both at the item level (e.g., the word "mommy" is likely to be easy in both Polish and English) and, more broadly, at the category level. Previous research has shown that certain semantic and lexical categories, such as sound-related words and nouns, tend to exhibit consistent levels of easiness across languages (e.g., Frank et al., 2021). We can expect that item difficulty estimates the relative ordering of items will remain correlated, particularly since both CAT-CDIs were derived from corresponding long-form CDIs and include partially overlapping items. Nevertheless, the CAT-CDIs do not need to show similarities across languages: for practical usefulness, it is more important that each instrument's item difficulties accurately reflect its own calibration sample. Even if parallel items differ across English and Polish, such differences do not undermine CAT performance, as long as the parameters yield precise and efficient ability estimates within each language.

CAT settings: ending an administration. Computerized adaptive testing involves numerous design choices, such as (but not limited to) how to estimate the initial ability estimate, which algorithm to use for estimating the ability during the task, and which criterion to apply when selecting the next item. In these aspects, the Polish CAT-CDIs followed the procedures described in detail in Kachergis et al. (2022). But one of the important decisions concerns the stopping criteria of the CAT task. These criteria can be based on a maximum number of administered items or reaching a target level of estimate precision – typically expressed as a standard error (SE) of the estimate. The English CAT-CDI prioritized a shorter CAT typically ending after 50 items, or sooner if a standard error (SE) of 0.15 was reached. In contrast, the Polish CAT-CDI prioritized greater precision, continuing until an SE of 0.1 was reached. If this level of precision could not be reached, the test concluded after a maximum of 75 items. These differences in CAT settings may result in relatively shorter English CAT-CDIs but relatively more precise estimates obtained from the Polish CAT-CDIs. Importantly, neither approach is inherently better; rather, each reflects different design priorities and assumptions about what a CAT-CDI should optimize - whether efficiency of administration, precision of measurement, or some appropriate balance between the two.

Present Study

Our aim is to evaluate the potential usefulness of CAT-CDIs for both research and clinical applications, using evidence from the validation studies of the two CAT-CDIs in English and in Polish. For a CAT-CDI to be effective - whether for screening, diagnosis, or broader research purposes - we propose that it should meet the following criteria:

1. **CAT-CDIs should require less administration time.** To this end, we will compare the median administration time between the full CDIs and CAT-CDIs.
2. **CAT-CDIs should demonstrate good concurrent validity with the full CDI and good reliability**. Concurrent validity will be evaluated via correlations between scores and estimates obtained from the full CDIs and CAT-CDIs. We will also explore cases of discrepancy between the two tools. We will asses precision by examining the standard error of the measurement across the ability range, and empirical reliability of the two CAT-CDIs, in comparison to the respective full CDIs.
3. **CAT-CDIs should also avoid reliance on the same subset of items across administrations**, to support potential for repeated use, e.g. in a longitudinal study

or progress monitoring. To examine this, we will calculate the frequency with which each item appeared across CAT-CDI administrations, expressed as a percentage of all administrations.

4. **CAT-CDIs across language may (but do not have to) show related item properties**. Previous research has shown that certain semantic and lexical categories, such as sound-related words and nouns, tend to exhibit consistent levels of easiness across languages (e.g., Frank et al., 2021). We will compare and correlate item difficulty estimates across the two languages, hypothesizing that although absolute difficulty values may differ due to different samples (due to differences in calibration samples), the relative ordering of items will remain correlated, particularly within categories that show robust cross-linguistic consistency.

We will report these outcomes in the two CAT-CDIs, in English and in Polish and discuss similarities and differences in the context of how the development of two CAT-CDIs differed. We aim for these analyses to not only assess the validity of the CAT-CDIs but also provide insights that can guide future adaptations in their core developmental choices.

Methods

Polish validation study

Participants. In the Polish validation study, 147 parents from an ongoing study were invited via email to take part in the validation study. The sample was White (reflecting the limited ethnic diversity within the Polish population). However, there were 28 children multilingual with Polish (as their home language) and some other (societal) language, living outside of Poland. We considered this group to be too large to exclude from the study, but not large enough to warrant a separate set of analyses. However, we included some comparisons between the Polish monolingual and bilingual groups in Supplementary Material. Next, we excluded participants for whom the time between full

CDI and CAT testing was longer than 30 days (n = 31), children who were out of the CDI:WS age range when parents filled in either full or CAT CDI (n = 3). The final dataset included 113 parents (49 girls, 43.36%). The children were aged between 18 and 34 months ($M = 24.73$, $Mdn = 24$, SD = 4.47). Parents were asked about their education level and 65 (57.52%) reported having higher education or PhD, 1 (0.88%) reported having unfinished higher education, 11 (9.73%) reported having secondary education and 36 (31.86%) did not report their education level.

Materials. The data presented here was gathered with CAT-CDIs in Polish presented above. It includes 666 items, of which 395 words (58%) are also found in the American English CAT-CDI.

Procedure. Polish parents completed the full CDI and the CAT-CDI online using a web app (CDI-Online) (Krajewski et al. 2020) designed specifically for CDI administration. Since the CAT-CDI was added as a final step to an ongoing study, the order of the CDIs was fixed, with the full CDI always first. After 1–30 days ($M = 8.31$, $Mdn = 8$, SD = 4.90) parents were asked to fill in the CAT-CDI. In the full CDI, each page presents all items in a given semantic category. The smallest categories (with fewest items on a page) include under 10 items, while the biggest category, action words, contains over 100 items. In CAT-CDI, parents are presented with one word at a time and asked to select one of two answers “Yes” or “No” to indicate whether their child says that word. The word is accompanied by information on the semantic category it belongs to.

American English validation study

The Polish data is hereby contrasted with the American English data from the validation study reported before in Kachergis et al. (2022). The final dataset included 204 children (103 girls, 50.49%). The children were aged between 15 and 36 months ($M = 26.21$, $Mdn = 26$, SD = 6.37). The average number of years of education attained by the primary caregiver was 15.82 (SD = 2.27).

The CAT-CDI was administered within the Web-CDI interface (DeMayo et al. 2021). Parents were given both CDIs in one sitting, but the order of the CDI versions was counterbalanced. As in the Polish CDIs, the view of the CDI differs depending on whether it is a full CDI or a CAT-CDI. In the full CDI, each page presents all items in a given semantic category (with categories ranging from under 10 to over 100 items). In CAT-CDI in both languages, parents are presented with one word at a time and asked to select one of two answers "Yes" or "No" to indicate whether their child says that word. The word is accompanied by information on the semantic category it belongs to.

Results

In the following section we will explore the two CAT-CDI's performance on those criteria. The Polish data has not been reported previously. The English has been reported in Kachergis et al. (2022) and was re-analysed for the purpose of the current paper.

Do CAT-CDIs require less time to administer?

CAT-CDIs were significantly shorter to administer: the English CAT-CDI median time was 342s (~5.7 minutes) (median full CDI time was 1466s (~24.43 minutes)) and the Polish CAT-CDI median time was 117s (~1.95 minutes) (median full CDI time was 1240s (~20.67 minutes)). Even though the CAT-CDI in Polish seemed shorter (compared to English CAT-CDI), the number of items administered in the two language versions of CAT-CDI was comparable: English $M = 31.42$, $Mdn = 25$ (25–50 items), Polish $M = 34.71$, $Mdn = 23$ (13–75 items). Notably, as the medians indicate, for half of the participants, the estimate precision was obtained relatively early and the CATs ended after 25 (English) or 23 (Polish) items.

Do CAT-CDIs demonstrate good concurrent validity and reliability?

Concurrent validity. To examine the concurrent validity of the two CAT-CDIs, we calculated correlations between CAT-CDI estimates and CDI raw scores. In both languages, we found similarly strong correlations between the abilities estimated from CAT-CDI and the full CDI scores (English and Polish: $r = .86$). The correlations were also strong within individual age groups (see Tables 1 and 2).

For each instrument, we also obtained IRT ability estimates from the full CDI using the same 2-PL model. Because these estimates draw on responses to all available items in the full CDI, they incorporate more information about each child's ability, reducing random error and increasing precision. Such ability estimates from the full test version are often considered a baseline for comparisons with ability estimates from the CAT version. We correlated the abilities estimated from the CAT-CDI and abilities estimated from full CDI and found strong correlations in both languages, English and Polish: $r = .92$. The relatively higher correlation indicates that CAT-CDI estimates align more closely with the latent ability captured by the full IRT model than with raw sumscores. Since the CAT–raw-score correlations were lower ($r = .86$), this suggests that raw CDI sumscores are less tightly linked to the underlying ability and may therefore yield lower validity than IRT-based scoring.

Finally, to examine consistency between model-based and traditional scores, we correlated the abilities estimated from the full CDI and the full CDI scores, i.e. two measures obtained from the same CDI version. Again, we found strong correlations in both languages (English: $r = .94{=}5$, Polish: $r = 0.94$).

Table 1

English: Correlations between ability estimated by CAT-CDI and ability estimated from full CDI by children's age.

	[15,18)	[18,21)	[21,24)	[24,27)	[27,30)	[30,33)	[33,36]
r ability CAT vs full CDI	0.95	0.85	0.82	0.83	0.59	0.84	0.86
N	26	22	26	30	28	24	48

Table 2

Polish: Correlations between ability estimated by CAT-CDI and ability estimated from full CDI by children's age.

	[15,18)	[18,21)	[21,24)	[24,27)	[27,30)	[30,33)	[33,36]
r ability CAT vs full CDI	NA	0.8	0.94	0.91	0.89	0.95	NA
N	NA	29	22	16	23	22	1

Discrepancy. Next, we checked for the difference between the abilities as estimated by CAT-CDI and the full CDI. For each participant, we calculated the mean squared error (MSE): we subtracted the CAT-CDI estimate from the full estimate and squared this difference to eliminate negative values. The mean squared error in English was 0.57 (Mdn = 0.35, SD = 0.70), and in Polish it was 0.19 (Mdn = 0.08, SD = 0.45). The MSE was relatively high in English (compared to Polish), indicating a relatively larger discrepancy between CAT-CDI ability estimates and full CDI ability estimates in English. But in both languages, the difference in estimates occurred despite strong correlations between the two measures. This difference between the full and CAT-CDI estimates may be explained by at least two reasons. First, a number of participants with extreme discrepancies between the CAT-CDI and full CDI estimates may be inflating the MSE.

Second, there might be a systematic bias, with the CAT estimate being higher than the full CDI estimate either consistently across the ability range or within a specific segment of it.

To explore the first of these possibilities, we identified children for whom the estimates from the CAT-CDI and full CDI showed extreme discrepancies, i.e. the difference between the two estimates was removed by 1.5 SD from the mean. There were 15 such cases (7.35%) in the English dataset and 4 cases (1.96%) in the Polish dataset. We then re-calculated the mean squared error without these cases, which yielded an MSE of 0.44 (Mdn = 0.29, SD = 0.44) in English and 0.12 (Mdn = 0.07, SD = 0.14) in Polish. The updated MSE values were only slightly lower than the original ones, suggesting that extreme discrepancies had only a modest effect on the overall error and cannot fully account for the differences between ability estimates from the full CDI and the CAT-CDI in the two languages.

Next, we examined a possible bias in our two ability estimates. We found that the mean CAT ability estimates were higher than the full CDI ability estimates. That was true both in English, $M_D = 0.55$, 95% CI $[0.48, 0.62]$, $t(203) = 15.11$, $p < .001$ (large effect size: Cohen's $d = 1.06$, 95% CI $[0.89, 1.23]$), and in Polish, $M_D = 0.26$, 95% CI $[0.19, 0.32]$, $t(112) = 7.71$, $p < .001$ (medium effect size: Cohen's $d = 0.73$, 95% CI $[0.52, 0.93]$). A closer look showed that the difference between the full and CAT-CDI estimates depended on the child's ability level (see Figure 1). For low ability, the differences between the estimates were larger, and the English CAT-CDI estimates tended to be higher than the full scores. For the highest ability, the estimates from the two CDIs seemed most aligned. In Polish, the CAT-CDI estimates generally aligned more closely with the full CDI estimates across the whole range of ability. Still, a similar trend has emerged as in English, though weaker: for the lowest ability levels, the CAT-CDI estimates were mostly higher than the full CDI estimates. However, for the highest ability levels, the CAT-CDI estimates were in fact lower than the full CDI estimates.

The question that remains is whether the differences between the two estimates

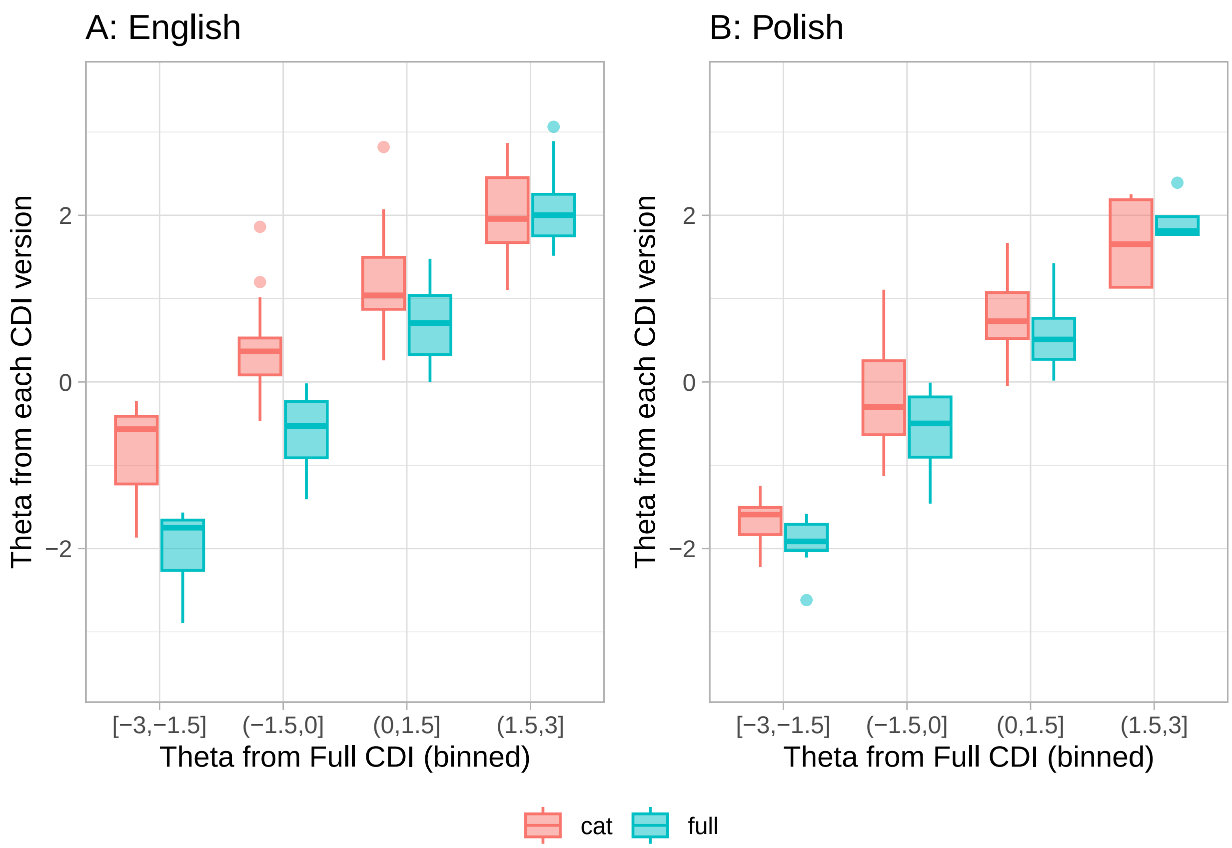


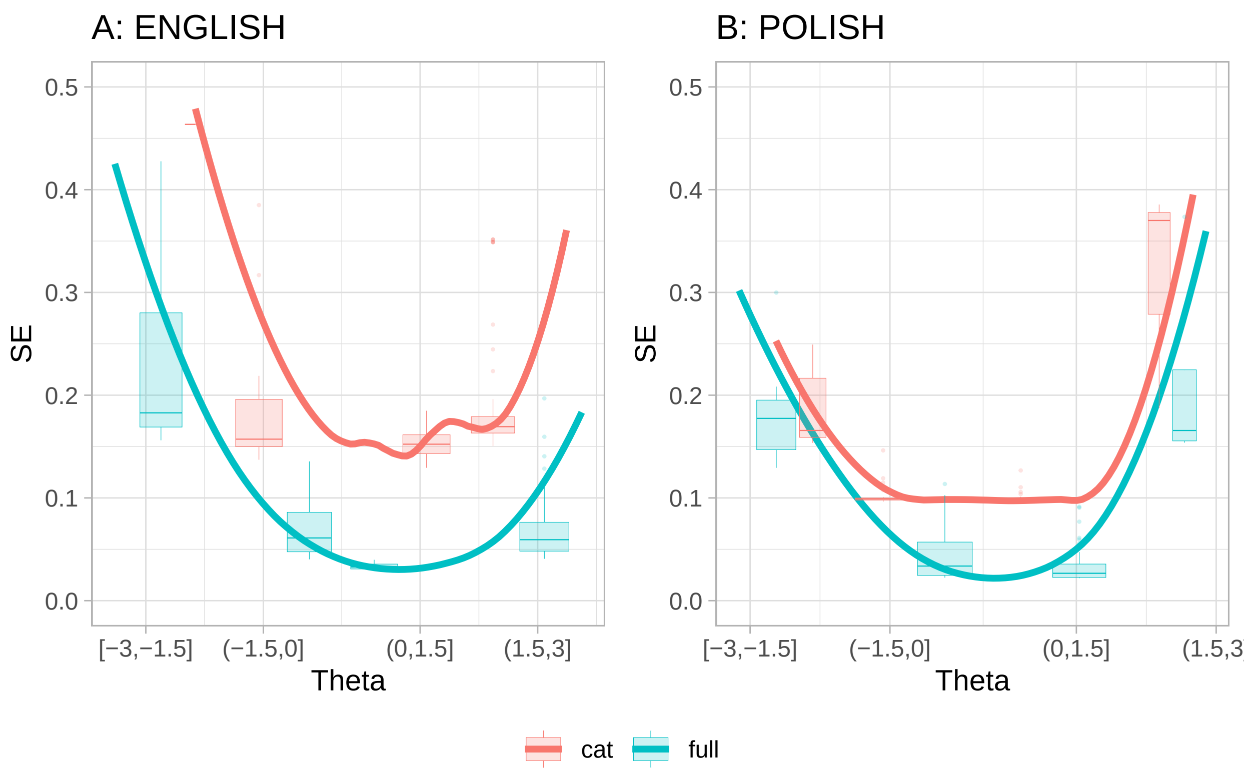
Figure 1. The relation between ability (estimated from full CDI) and the size of discrepancy between the estimates from full CDI and CAT-CDI: (A) English, (B) Polish.

(visible especially in the lower ability, as in Figure 1) come from CAT over-estimating child's true lexical ability or from the full CDI underestimating child's ability. For one, it may be that in CAT-CDI parents answer "yes, my child produces this word" to more items that would be expected (Kachergis et al. 2022), and as a result, CAT-CDI may overestimate the child's true ability. On the other hand, it could be that as the full CDI is very long, parents may omit marking some items that their child produces (see e.g., Łuniewska et al., 2025 for parental inconsistencies in reporting vocabulary of older children). This could result in the full CDI underestimating the child's true ability. It is difficult to answer this question without direct assessment of child's lexical ability. With the existing data, we can at least compare the precision of the two estimates to assess which measure is likely more reliable.

Precision and reliability of the estimates. We looked at how precision of the estimate, measured by standard error of the theta (SE), related to the ability levels (see Figure 2). In both languages, the characteristic U-shaped curves indicate that measurement precision is the highest around the middle of the ability range (i.e., showing lowest SE) and decline at the extremes (i.e., for very low or very high ability the SE values are the highest). This U-shape is observed for both the CAT-CDI and the full CDI, indicating that even the full CDI provides relatively less precise estimates at the extreme ends of ability (i.e. floor and ceiling scores). Note that in Polish (Figure 2, panel B) the lowest ability estimates (i.e., [-3,-1.5]) yielded a slightly lower median SE in CAT-CDI than in the full CDI, suggesting that at lowest ability levels, CAT-CDI might be similarly or slightly more precise than the full CDI. But for the highest ability estimates (i.e., (1.5, 3]) it was the full CDI that came with better precision than the CAT-CDI.

Nevertheless, in both CAT-CDIs, we have been able to reliably estimate lexical ability for the majority of children. According to Makransky, a standard error (SE) below 0.2 is considered acceptable. In practical terms, an SE of 0.2 indicates that the estimated ability is likely within +/- 0.2 units of the child's true lexical ability. Such precision of the CAT-CDI estimate was obtained for 189 out of 204 American participants (92.60%) and 109 out of 113 Polish participants (96.50%). The remaining participants were at the extremes of the ability scale (see Figure 2). The ability estimates from the Polish CAT-CDI generally fell well below an SE of 0.1, except for the highest-ability observations. Because the Polish CAT-CDI was designed to stop once an SE of 0.1 was reached, the precision of Polish estimates never went below 0.1 (i.e., the CAT-CDI was stopped at this point).

We also examined the empirical reliability of the two CAT-CDIs, in comparison to the respective full CDIs. The reliability of the English CAT-CDI was 0.95 (for comparison, the full CDI: 1.00). The reliability of the Polish CAT-CDI was 0.98 (for comparison, the full CDI: 0.99). Thus, the empirical reliability values for both the CAT-CDIs were excellent.



Note that in English there is only one case where CAT estimate was smaller than −1.5, therefore no CAT boxplot is plotted in that bin.

Figure 2. The relation between ability estimates from full CDI and CAT-CDI and the standard error of that estimate: (A) English, (B) Polish.

Do CAT-CDIs avoid reliance on the same subset of items?

It is to be expected that a CAT-CDI will not need to administer all the items from the CDI item bank. In fact, in all the task administrations in the validation study, the English CAT-CDI used altogether 251 items (36.91%) and the Polish CAT-CDI used altogether 258 items (38.74%) from their respective item banks. By design, a CAT-CDI selects items that provide the most information for each participant. In practice, this means CAT draws from a subset of items that both match the participant's current ability estimate in difficulty and show high discrimination, allowing clear distinctions between individuals with similar ability levels. A potential concern, however, is that a CAT-CDI might repeatedly rely on the same subset of optimal items. CAT-CDI's usefulness in research—particularly in longitudinal studies—lies in the possibility of administering the CDI frequently without parents becoming overly familiar with specific, frequently repeated items. This advantage holds only if the CAT-CDI does not consistently present the same items across administrations. Therefore, we investigated whether the two CAT-CDIs repeatedly selected any items too frequently. In English, 3 items appeared in more than half of administrations - *"long"* in 64% of administrations, *"make"* in 54% of administrations, and *"last"* in 53% of administrations. As expected, they are also high discrimination items in their difficulty bins. "Long" is additionally one of the starting items, i.e. a first item shown to the parent, chosen so that there is high probability the parent can respond positively. In Polish, 3 items appeared in more than half of administrations - *"szukać"* (to look for) appearing in 83% of administrations, *"znaleźć"* (to find) appearing in 61% of administrations, and *"babcia"* (grandmother) appearing in 60% of administrations. As expected, all three items show very high discrimination. *"Szukać"* (to look for) and *"znaleźć"* (to find) are in fact in top 5 discrimination items in general. *"Babcia"* (grandmother) is one of the starting items. The fact that only 3 items out of c.a. 250 (c.a. 1%) are used in more than half of administrations is to CAT-CDI's advantage.

Do CAT-CDIs show related item difficulty across languages?

There are 679 items in the English CAT-CDI and 666 items in the Polish CAT-CDI. Of these, 431 are translation equivalents, i.e. items whose meaning is common across the two languages (e.g. "cat", "kot"). The translation equivalents were identified by using "universal lemmas", or "unilemmas", i.e., cross-linguistic conceptual mappings that link words with broadly similar meanings across languages (constructed and verified using Wordbank glosses) to enable systematic comparison of CDI items (Kachergis et al., n.d.).

The items' difficulty and discrimination parameters were calculated using IRT 2 parameter model (these included separate samples, see Kachergis et al. 2022 and Krajewski et al., in preparation). An item's difficulty indicates the ability level at which there is a

50% probability that a participant will respond correctly (here: that a child will know a word). Difficulty is expressed on the same scale as theta (measured ability): negative values correspond to harder items, values near zero to items of moderate difficulty, and positive values to easier items (note that difficulty is inversely related to theta and so, negative theta means low ability while positive theta means high ability). An item's discrimination shows how well it can distinguish between individuals whose abilities are close. In this paper, we focus mainly on item difficulty, as it is directly tied to the ability being measured, whereas discrimination reflects more the quality of the item as a measurement tool.

English items were in general more difficult than the Polish items (English: min = -7.16, max = 4.45, M = -1.75, Mdn = -1.83, SD = 2.06; Polish: min = -4.34, max = 4.41, M = -0.32, Mdn = -0.43, SD = 1.41). This was true also for a subset of 431 items common to both languages - English items still proved to be more difficult than the Polish ones: $M_D = -1.82$, 95% CI $[-1.96, -1.68]$, $t(430) = -25.38$, $p < .001$. This is a result of the characteristics of the samples used to estimate the item parameters. In English, item difficulty was calculated based on a broader sample of children aged 12–36 months, spanning the CDI:WG, CDI:WS, and CDI-III (reflecting the idea of having one CAT-CDI for all the age versions), whereas the Polish data came from a sample of narrower age range of 18–36 months, corresponding to the CDI:WS. As a result, item difficulty in English was estimated using a relatively younger sample, for whom certain items may have been more challenging – thus appearing more difficult – compared to the older Polish sample. Still the item difficulty for common items in the two languages was positively and moderately correlated: $r = .66$, 95% CI $[.60, .71]$, $t(429) = 18.12$, $p < .001$.

We also examined whether CDI semantic categories show comparable difficulty values across the two languages. To this end, we conducted a series of Spearman's rank correlations on item difficulty within each semantic category (we considered only items common to both the Polish and English CDIs) (see Table 3). The rank correlations coefficients varied by category, but for half of the categories the correlations were moderate to strong (0.51 to 0.91).

The least correlated semantic category was Quantifiers. The discrepancy may be partially to the fact that many English quantifiers map onto multiple Polish equivalents - for example, “some” maps to two Polish CDI items, “jeszcze” (with a similar difficulty value as the English “some”) and “więcej” (more difficult than “jeszcze” and “some”). These two Polish items are used in different contexts, i.e., “jeszcze” would be used more often in child directed speech than “więcej” also because it can be used to refer to both quantity and repetitive actions (meaning “again”). “Więcej” implies a quantitative comparison of some sorts, as in “more than before” or “more than two”.

Interestingly, the second least correlated semantic category were Clothes. Here, some divergence may be explained by one-to-multiple mapping (e.g., “hat” is either “czapka” or “kapelusz” in Polish), but also words characteristics, e.g., “bib” in Polish is a much more complex word, “śliniaczek”; English “underpants” are more morphologically complex (and maybe less frequent) than Polish “majtki”.

Taken together, these two semantic categories reflect the existing cross-linguistic differences in lexical mapping as well as item-specific factors such as frequency, morphological complexity, and contextual usage.

Table 3

Cross-linguistic correlations of item difficulty within each CDI semantic category (Spearman's rank correlations).

CDI category	rho	n
quantifiers	0.11	10
clothing	0.20	20
connecting_words	0.23	8
locations	0.31	6
pronouns_demonstrative	0.32	4
places	0.33	8
vehicles	0.45	9
furniture_rooms	0.47	17
action_words	0.51	78
body_parts	0.52	17
descriptive_words	0.55	23
prepositions	0.55	10
time_words	0.57	8
outside	0.58	20
games_routines	0.62	18
household	0.63	30
food_drink	0.64	42
people	0.67	16
animals	0.68	29
pronouns_personal	0.72	26
toys	0.75	14
sounds	0.83	8
question_words	0.91	10

Last, we investigated whether particular lexical categories show related difficulty values across the two languages. We considered the following lexical classes: verbs, nouns, adjectives, function words and other, i.e. mostly sounds and words relating to routines (e.g. “hello”, “yes”, “thank you”) and time (e.g. “tomorrow”, “day”). Again, we performed a series of Spearman’s rank correlations (on the difficulty of items common to CDI-CATs in Polish and English) for each lexical class (see Table 4. The weakest correlation was for verbs (0.37), then function words (0.53), adjectives (0.53), nouns (0.61), other (0.75). Again, with verbs we saw that a single English item (e.g. “break”) can map onto two Polish items: “złamać” (to break by snapping or fracturing, usually used for solid objects) and “zepsuć” (to make something stop working properly). Because verbs are structurally tied to grammar (which is highly language-specific) they naturally vary in frequency in child-directed speech and in their communicative salience, which can affect how well children know them across languages. Notably, function words include many of the CDI semantic categories which showed the lowest correlations in Table 3, i.e., quantifiers, connecting words, locations, and pronouns. Again, many English function words map onto multiple Polish equivalents, e.g., “which” corresponds to “jaki” (when asking about type, kind, or quality) and “który” (when selecting from a limited set of options). These cross-linguistic differences partially explain the weaker correlation for function words. By contrast, the relatively high correlations for nouns, sounds and words relating to routines (grouped together as “other”) reflect their consistent easiness/difficulty across languages: words such as “mum”, “ball”, “hi”, “woof woof” are acquired early, aligning with findings that nouns are strongly and consistently over-represented in early vocabularies across many languages (e.g., Frank et al., 2021).

Table 4

lexical_class_en	rho	n
verbs	0.37	81
adjectives	0.53	25
function_words	0.53	76
nouns	0.60	197
other	0.74	52

Discussion

The MacArthur–Bates Communicative Development Inventories (CDIs) are widely used in both research and practice to assess children's early language abilities, particularly vocabulary knowledge. To ensure comprehensive coverage, CDIs contain extensive word lists that vary in difficulty, frequency, and lexical class. Because of their length, numerous efforts have been made to shorten CDI administration while preserving the tool's strong psychometric properties.

This paper presents validation data for shortened CDIs developed within the framework of Item Response Theory (IRT) and using Computerized Adaptive Testing (CAT). Specifically, we introduce two CAT-based CDIs (CAT-CDIs): one in American English and one in Polish. Although both were created using a broadly similar procedure, certain design choices differed across the two versions. For English, a single CAT-CDI was developed to measure word production across the age ranges covered by all three CDI versions (Words and Gestures, Words and Sentences, and CDI-III). In Polish, a separate CDI was created for each of those age versions. The English CAT-CDI prioritized efficiency, administering up to 50 items but stopping earlier if the target level of precision (a standard error of 0.2) was reached before the 50th item. The Polish CAT-CDI emphasized precision, allowing administrations of up to 75 items and targeting a smaller standard error (0.1). These differences highlight key decision points for researchers or clinicians aiming to develop a CAT-CDI in their own language.

The goal of the present study was to evaluate the usefulness of CAT-CDIs in both research and clinical contexts. Parents (in the USA and in Poland) were asked to fill in both a full version of the CDI and the CAT-CDI. From the CAT-CDI, we obtained the children's ability estimates (and their precision). From the full CDIs we obtained the raw scores and we also calculated the IRT estimates (and their precision). These estimates based on the responses to all items in a full CDI allowed us to obtain a baseline for comparisons with ability estimates from the CAT version. For a CAT-CDI to be considered useful, we propose that it should require less administration time while showing good validity and reliability. We also suggest that CAT-CDIs should avoid reliance on the same subset of items, to support potential for repeated use. But on the other hand, we do not expect the same item properties in the English and Polish CAT-CDIs, which we believe could reflect potential cross-linguistic effects. Below, we summarize and discuss our findings with respect to each of these criteria.

CAT-CDIs require less administration time. Both the English and the Polish CAT-CDIs proved much shorter than their full counterparts, with median times of 342s (~5.7 minutes) (English) and 117s (~1.95 minutes) (Polish). In both CAT-CDIs a comparable number of items was administered, English *Mdn* = 25 (25–50 items), Polish *Mdn* = 23 (13–75 items). Notably, as the medians indicate, for half of the participants, the estimate precision was obtained relatively early and the CATs ended after 25 (English) or 23 (Polish) items. NOTE: ENG administration time to be doubled-checked, there are 27 kids with CAT administration time in weeks (possibly an tab opened for too long). Karolina will discuss with George K.

CAT-CDIs show good validity and reliability. The CAT-CDI estimates correlate strongly with both full CDI scores and estimates obtained from the full CDI, proving high concurrent validity. The CAT-CDIs were strongly correlated with the full CDI versions in both languages, which supports the overall validity of the CAT approach. These correlations held for both monolingual and Polish multilingual participants (see Supplementary Materials), however, the Polish multilingual sample was small, so further research is needed to confirm these findings. The CAT-CDI estimates also showed excellent reliability. CAT-CDI scores diverged notably from full CDI estimates for only a small number of children in both language groups. A closer look showed that for the lower ability, the CAT-CDI estimates were higher than those of the full CDI, but in the Polish data, the precision of those CAT-CDI estimates was similar or slightly higher than that of the full CDI. This might potentially suggest that for children with lower lexical abilities, CAT estimates may be more aligned with the true abilities. For the higher ability, CAT estimates were more closely aligned with the full CDI estimates, but the precision of the CAT estimates was similar or lower than that of the full CDI. Importantly, for the large majority of children, i.e. 92.60% of the American participants and 96.50% of the Polish participants, the obtained CAT-CDI estimates were below the precision cut-off point suggested by Makransky et al. of 0.2.

CAT-CDIs avoid repeated reliance on the same subset of items. By design, a CAT-CDI selects items that provide the most information for each participant. In practice, this means participant of different abilities may see largely a different set of items. A potential concern, however, is that a CAT-CDI might repeatedly rely on the same subset of optimal items, i.e. items of matching difficulty and high discrimination. As a CAT-CDI's usefulness lies also in the possibility of administering the CDI frequently without parents becoming overly familiar with specific items, we investigated whether the two CAT-CDIs repeatedly selected and items too frequently. We found that only three items per language were administered to more than half of participants - each showing high discrimination - and this minimal overlap (about 1% of the item pool) highlights an efficiency advantage of the CAT-CDI.

CAT-CDIs across languages do not have to show similar item properties. The comparison of item parameters across English and Polish highlights both methodological and linguistic influences on CAT-CDI development. Differences in calibration samples, particularly the broader age range in English versus the narrower one in Polish, likely contributed to observed shifts in item difficulty, with English items tending to appear more difficult (this was true also for the subset of items found on the CDI lists in both languages). Still the item difficulty for common items in the two languages was positively and moderately correlated. At the same time, variation in lexical mapping, morphological complexity, and frequency of use may also shape item properties across languages. These findings underscore the importance of considering both sample characteristics and language-specific features when interpreting item parameters or developing new CAT-CDIs. Importantly, CAT-CDIs do not need to be identical across languages, since the primary requirement is that each instrument's item difficulties accurately reflect its own calibration sample. Differences in parallel items across English and Polish are both expected and scientifically interesting, as cross-linguistic comparisons of item properties are frequently investigated; nonetheless, such differences do not compromise CAT performance, as long as the parameters enable precise and efficient ability estimation within each language.

Conclusions

Overall, the two validation studies reported here indicate that the English and Polish CAT-CDIs can substantially reduce administration time while still providing reliable and valid estimates of children's early vocabulary. Both instruments showed strong correlations with full CDI measures, although some systematic differences in estimates were observed, particularly at the lowest and highest ends of ability, where the full CDI itself may also provide less precise estimates. The adaptive format also appeared to minimize over-reliance on a narrow set of items (as compared to a short CDI), supporting its potential for repeated use. At the same time, cross-linguistic comparisons highlighted that differences in calibration samples can affect item parameters, underscoring the need for careful consideration when adapting CAT-CDIs across languages. Importantly, the exact performance of a CAT-CDI is shaped by key design choices, such as the structure of the calibration procedures and stopping criteria. This suggests that future adaptations may need to weigh efficiency against precision to optimize the tool for their specific research or clinical goals.

References

Belteki, Z., Ward, E. K., Begum-Ali, J., van den Boomen, C., Bölte, S., Buitelaar, J., Charman, T., Demurie, E., Falck-Ytter, T., Hunnius, S., Johnson, M. H., Jones, E. J. H., Oosterling, I., Pasco, G., Pijl, M. K. J., Radkowska, A., Rudling, M., Tomalski, P., Warreyn, P., … Haman, E. (2025). A concurrent validity study of the Mullen Scales of Early Learning (MSEL) and the MacArthur-Bates Communicative Developmental Inventory (CDI) in infants with an elevated likelihood or diagnosis of autism. *J. Autism Dev. Disord.*, 1–16.

Bohn, M., Prein, J., Engicht, J., Haun, D., Gagarina, N., & Koch, T. (2025). PREVIC: An adaptive parent report measure of expressive vocabulary in children between 3 and 8 years of age. *Behav. Res. Methods*, *57*(3), 95.

Bohn, M., Prein, J., Koch, T., Bee, R. M., Delikaya, B., Haun, D., & Gagarina, N. (2024). oREV: An item response theory-based open receptive vocabulary task for 3- to 8-year-old children. *Behav. Res. Methods*, *56*(3), 2595–2605.

Chai, J. H., Lo, C. H., & Mayor, J. (2020). A Bayesian-Inspired Item Response Theory–Based Framework to Produce Very Short Versions of MacArthur–Bates Communicative Development Inventories. *Journal of Speech, Language, and Hearing Research*, *63*(10), 3488–3500. https://doi.org/10.1044/2020_JSLHR-20-00361

Dale, P. S. (1991). The Validity of a Parent Report Measure of Vocabulary and Syntax at 24 Months. *Journal of Speech, Language, and Hearing Research*, *34*(3), 565–571. https://doi.org/10.1044/jshr.3403.565

Ezeizabarrena, M.-J., Garcia Fernandez, I., & Murciano, A. (2024). Short versions of the Basque MacArthur-Bates Communicative Development Inventories (children aged 8–50 months). *Front. Commun.*, *9*, 1406829.

Fenson, L., Pethick, S., Renda, C., Cox, J. L., Dale, P. S., & Reznick, J. S. (2000). Short-form versions of the MacArthur Communicative Development Inventories. *Appl. Psycholinguist.*, *21*(1), 95–116.

Frank, M. C., Baumgartner, H. A., Braginsky, M., Kachergis, G., Lightbody, A. A., Sparks, R. Z., Zhu, R., Carlson, S. M., Graham, S., Lipina, S. J., Newcombe, N. S., Odgers, C. L., Pianta, R. C., Siegler, R. S., Snowling, M., Yoshikawa, H., Cubillo, A., & Dodge, K. A. (2025). Learning Variability Network Exchange (LEVANTE): A global framework for measuring children's learning variability through collaborative data sharing. *Child Dev.*, *cdev.70011*.

Frank, M. C., Braginsky, M., Yurovsky, D., & Marchman, V. A. (2021). *Variability and Consistency in Early Language Learning: The Wordbank Project*. MIT Press.

Garmann, N. G., Hansen, P., Simonsen, H. G., & Kristoffersen, K. E. (2019). The Phonology of Children's Early Words: Trends, Individual Variation, and Parents' Accommodation in Child-Directed Speech. *Front. Commun.*, *4*.

Gatt, D., O'Toole, C., & Haman, E. (2015). Using Parental Report to Assess Early Lexical Production in Children Exposed to More Than One Language. In S. Armon-Lotem, J. de Jong, & N. Meir (Eds.), *Assessing Multilingual Children* (pp. 151–195). Multilingual Matters. https://doi.org/10.21832/9781783093137-009

Kachergis, G., Marchman, V. A., Dale, P. S., Mankewitz, J., & Frank, M. C. (2022). Online Computerized Adaptive Tests of Children's Vocabulary Development in English and Mexican Spanish. *Journal of Speech, Language, and Hearing Research*, *65*(6), 2288–2308. https://doi.org/10.1044/2022_JSLHR-21-00372

Kachergis, G., Tan, A. W. M., Marchman, V. A., Dale, P., & Frank, M. C. (n.d.). *Measuring children's early vocabulary in low-resource languages using a swadesh-style word list*. https://doi.org/10.31234/osf.io/njm7d_v1

Kim, S. W., Jeon, H. R., Park, E. J., Kim, H. I., Jung, D. W., & Woo, M. R. (2014). The Usefulness of M-B CDI-K Short Form as Screening Test in Children With Language Developmental Delay. *Annals of Rehabilitation Medicine*, *38*(3), 376–380. https://doi.org/10.5535/arm.2014.38.3.376

Lasorsa, F. R., Caselli, M. C., Simonelli, I., Stefanini, S., Pasqualetti, P., & Rinaldi, P.

(2021). Exploring Agreement between MB-CDI Short Forms for Evaluating the Language Skills of Italian Children Aged 18–24 Months. *Folia Phoniatrica Et Logopaedica*, *73*(6), 552–564. https://doi.org/10.1159/000513243

Lemhöfer, K., & Broersma, M. (2011). Introducing LexTALE: A quick and valid Lexical Test for Advanced Learners of English. *Behavior Research Methods*, *44*(2), 325–343. https://doi.org/10.3758/s13428-011-0146-0

Ma, W. A., Richie-Halford, A., Burkhardt, A. K., Kanopka, K., Chou, C., Domingue, B. W., & Yeatman, J. D. (2025). ROAR-CAT: Rapid Online Assessment of reading ability with computerized adaptive testing. *Behav. Res. Methods*, *57*(1), 56.

Makransky, G., Dale, P. S., Havmose, P., & Bleses, D. (2016). An Item Response Theory–Based, Computerized Adaptive Testing Version of the MacArthur–Bates Communicative Development Inventory: Words & Sentences (CDI:WS). *Journal of Speech, Language, and Hearing Research*, *59*(2), 281–289. https://doi.org/10.1044/2015_JSLHR-L-15-0202

Mayor, J., & Mani, N. (2019). A short version of the MacArthur–Bates Communicative Development Inventories with high validity. *Behav Res*, *51*(5), 2248–2255.

O' Toole, C., & Fletcher, P. (2010). Validity of a Parent Report Instrument for Irish-speaking Toddlers. *First Lang.*, *30*(2), 199–217.

Smoczyńska, M., Krajewski, G., Łuniewska, M., Haman, E., Bulkowski, K., & Kochańska, M. (2015). *Inwentarze Rozwoju Mowy i Komunikacji (IRMIK): SŁOWA I GESTY, SŁOWA I ZDANIA. Podręcznik*. IBE.

Smolík, F., & Bytešníková, I. (2021). Validity of the SDDS: A 40-item vocabulary screening tool for 18- to 42-month olds in Czech. *Journal of Communication Disorders*, *93*, 106146. https://doi.org/10.1016/j.jcomdis.2021.106146